Finite-temperature effects in enzymatic reactions — Insights from QM/MM free-energy simulations

Hans Martin Senn, Johannes Kästner, Jürgen Breidung, and Walter Thiel

Abstract: We report potential-energy and free-energy data for three enzymatic reactions: carbon–halogen bond formation in fluorinase, hydrogen abstraction from camphor in cytochrome P450cam, and chorismate-to-prephenate Claisen rearrangement in chorismate mutase. The results were obtained by combined quantum mechanics/molecular mechanics (QM/MM) optimizations and two types of QM/MM free-energy simulations (free-energy perturbation and umbrella sampling) using semi-empirical or density-functional QM methods. Based on these results and our previously published free-energy data on electrophilic substitution in *para*-hydroxybenzoate hydroxylase, we discuss the importance of finite-temperature effects in the chemical step of enzyme reactions. We find that the entropic contribution to the activation barrier is generally rather small, usually of the order of 5 kJ mol^{-1} or less, consistent with the notion that enzymes bind and pre-organize the reactants in the active site. A somewhat larger entropic contribution is encountered in the case of chorismate mutase where the pericyclic transition state is intrinsically more rigid than the chorismate reactant (also in the enzyme). The present results suggest that barriers from QM/MM geometry optimization may often be close to free-energy barriers for the chemical step in enzymatic reactions.

Key words: QM/MM, combined quantum mechanics/molecular mechanics, free-energy methods, free-energy perturbation, enzyme reactions.

Résumé : On rapporte les données d'énergie potentielle et d'énergie libre pour trois réactions enzymatiques, soit la formation d'une liaison carbone-halogène avec la fluorinase, l'enlèvement d'un hydrogène à partir du camphre avec le cytochrome P450cam et un réarrangement de Claisen chorismate à préphénate avec une chorismate mutase. Les résultats ont été obtenus par des optimisations MQ/MM semi-empiriques ou de la théorie de la fonctionnelle de la densité (TFD) et deux types de simulations d'énergie libre MQ/MM (perturbation de l'énergie libre et échantillonnage parapluie). Sur la base de ces résultats et des résultats d'énergie libre publiés antérieurement pour la substitution électrophile avec l'hydroxylase de *p*-hydroxybenzoate, on discute de l'importance d'effets de température finie sur l'étape chimique des réactions enzymatiques. Il a été observé que la contribution entropique à la barrière d'activation est relativement faible, généralement inférieure à 5 kJ mol^{-1}, en accord avec la notion que les enzymes se lient et organisent les réactifs dans le site actif. On rencontre une contribution entropique un peu plus importante dans le cas de la chorismate mutase pour laquelle l'état de transition péricyclique est intrinsèquement plus rigide que le réactif chorismate, aussi dans l'enzyme. Les résultats présentés ici suggèrent que les barrières obtenues à partir de l'optimisation MQ/MM de la géométrie se rapprochent souvent des barrières d'énergie libre pour l'étape chimique des réactions enzymatiques.

Mots-clés : MQ/MM, combinaison de mécanique quantique et de mécanique moléculaire, méthodes de l'énergie libre, perturbation de l'énergie libre, réactions enzymatiques.

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Introduction

Combined quantum mechanics/molecular mechanics (QM/MM) approaches are the method of choice for treating chemical reactions in enzymes,[1–4] providing an efficient and accurate description of complex biomolecular systems. They allow detailed insights into the mechanisms of enzyme reactions; see, e.g., refs. 3–5 for recent overviews. The QM/MM method is essentially a potential-energy method. However, the key quantity to characterize chemical systems and proc-

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H.M. Senn,[1,2] J. Kästner,[3,4] J. Breidung, and W. Thiel.[5] Max-Planck-Institut für Kohlenforschung, D-45470 Mülheim an der Ruhr, Germany.

[1]Corresponding author (e-mail: senn@chem.gla.ac.uk).
[2]Present address: WestCHEM and Department of Chemistry, University of Glasgow, Glasgow G12 8QQ, UK.
[3]Corresponding author (e-mail: kaestner@theochem.uni-stuttgart.de).
[4]Present address: University of Stuttgart, D-70569 Stuttgart, Germany.
[5]Corresponding author (e-mail: thiel@mpi-muelheim.mpg.de).

esses is the free energy. The free-energy difference between two states is directly related to the probability of finding the system in one state or the other. For an *NVT* system with constant number of particles, *N*, constant volume, *V*, and constant temperature, *T*, the relevant free energy is the Helmholtz free energy, *A*,

$$[1] \qquad A = U - TS$$

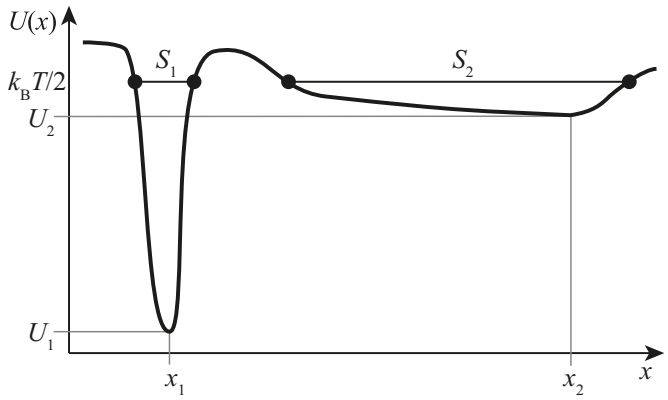
The free energy is composed of the internal energy, *U* (which is the potential energy plus the zero-point vibrational energy plus finite-temperature corrections, which are usually small), and the entropic contribution, –*TS*. At temperatures $T > 0$ K, knowledge of the energy alone is therefore not sufficient to determine the free energy. A higher-energy state with high entropy may be more favourable (that is, have a lower free energy) than a low-energy state with low entropy (see Fig. 1). The entropy can be related, loosely speaking, to the size of the configuration space available to the system at a given temperature.

In quantum chemistry, the conventional approach to computing the free energy of a molecule is based on the rigid-rotor/harmonic-oscillator (RRHO) approximation,[7] where the partition function is formulated under the assumptions that the molecule as a whole rotates as a rigid body and that all intramolecular motions can be treated as harmonic vibrations. However, when applied to large biomacromolecular systems, the RRHO treatment is problematic for several reasons: (*i*) A full vibrational analysis is computationally very costly or even prohibitive for systems typically encountered when modeling enzymatic reactions (several 1000 to 10 000 atoms). (*ii*) There are many "soft", anharmonic degrees of freedom in a (solvated) biomolecule for which the harmonic-oscillator approximation is inadequate. (*iii*) Even if the above points could be resolved, there remains a major issue: The RRHO approach is applied at a stationary point (minimum or transition state) obtained from a potential-energy minimization procedure. However, for large systems, this configuration will be just one of very many of similar energy, and one cannot assume it to be *the* representative structure.[8]

A more appropriate approach for biomolecular systems is therefore to calculate free energies from some type of sampling procedure at finite temperature that is able to explore, at least locally, the configuration space available to the system. The sampling can be done either with Monte Carlo or molecular-dynamics (MD) simulation techniques; we will restrict ourselves here to MD-based methods. The general idea is to express the free-energy difference between two states as a statistical-thermodynamical average that can be evaluated by sampling the appropriate ensemble (e.g., the canonical, or *NVT*, ensemble or the isothermal–isobaric, or *NPT*, ensemble).

Over the last few years, numerous studies have been reported that combine QM/MM methods with a variety of sampling techniques to calculate reaction and activation free energies of enzymatic processes; see refs. 3, 4, and 9 for recent reviews. In the present contribution, we have used QM/MM free-energy perturbation (FEP) as well as umbrella sampling (US) to investigate several enzyme reactions, namely, the halogenation step in fluorinase, the hydrogen-abstraction step in cytochrome P450cam, and the Claisen rearrangement in chorismate mutase. We will incorporate into the discussion previously reported free-energy results on other systems and obtained by other methods. From the combined body of data, we will attempt to draw more general conclusions on the effect of finite temperature on the chemical step of these, and perhaps other, enzymatic reactions.

Fig. 1. The high-energy state x_2 has a lower (more favourable) free energy than x_1 because of its higher entropy, indicated by the larger width of the potential well (adapted from ref. 6).



We will begin by briefly outlining our QM/MM method and discuss the QM/MM free-energy perturbation method, with which the majority of results herein were obtained. We will then introduce the three enzymatic reactions studied and provide computational details. The Results section will present potential-energy and free-energy data for the reactions, which will subsequently be discussed with a view on the importance of finite-temperature effects in the chemical step of enzyme reactions. We will finish with some conclusions.

Theoretical background

The QM/MM energy

Here, we outline our QM/MM scheme to facilitate the following discussion and introduce some notation. In brief, we use an additive scheme with hydrogen link atoms and electrostatic embedding with charge shifting.[3,10–13] The total (potential) energy of the entire system is written as

$$[2] \qquad E_{\mathrm{QM/MM}}(\mathbf{R}_{\mathrm{QM}},\mathbf{R}_{\mathrm{MM}}) = E_{\mathrm{QM}}(\mathbf{R}_{\mathrm{QM}};\mathbf{R}_{\mathrm{MM}}) + E_{\mathrm{MM}}(\mathbf{R}_{\mathrm{MM}}) + E_{\mathrm{QM-MM}}(\mathbf{R}_{\mathrm{QM}},\mathbf{R}_{\mathrm{MM}})$$

where $\mathbf{R}_{\mathrm{QM}}$ and $\mathbf{R}_{\mathrm{MM}}$ are the coordinates of the QM and the MM atoms, respectively; link atoms are included in $\mathbf{R}_{\mathrm{QM}}$ to simplify the notation. $E_{\mathrm{QM}}(\mathbf{R}_{\mathrm{QM}};\mathbf{R}_{\mathrm{MM}})$ is the electronic energy of the QM system in the presence of the MM point charges, but *without* accounting for the electrostatic interaction with the surrounding MM atoms (see below). $E_{\mathrm{MM}}(\mathbf{R}_{\mathrm{MM}})$ is the energy of the MM atoms as calculated by the classical force field. All QM–MM interactions are collected in $E_{\mathrm{QM-MM}}(\mathbf{R}_{\mathrm{QM}},\mathbf{R}_{\mathrm{MM}})$:

$$[3] \qquad E_{\mathrm{QM-MM}} = E^{\mathrm{b}}_{\mathrm{QM-MM}} + E^{\mathrm{vdW}}_{\mathrm{QM-MM}} + E^{\mathrm{el}}_{\mathrm{QM-MM}}$$

$E^{\mathrm{b}}_{\mathrm{QM-MM}}$ and $E^{\mathrm{vdW}}_{\mathrm{QM-MM}}$ contain the bonded (typically bond stretching, angle bending, torsion) and van der Waals interactions between the QM and the MM atoms, respectively. These terms are calculated at the force-field level. The elec-

trostatic QM–MM interaction, E^{el}_{QM-MM}, is treated within the electrostatic embedding scheme, that is, the QM electron density is polarized by the MM charges. This is achieved by including the point charges into the QM Hamiltonian, which amounts to solving the SCF equations in the presence of the point-charge field. The corresponding electronic Hamiltonian thus reads (in atomic units)

$$\begin{aligned}\widehat{H}_{SCF} &= \widehat{H}_0 + \widehat{H}^{el}_{QM-MM} \\ &= \widehat{H}_0 - \sum_i^{\text{electrons}} \sum_k^{\text{point charges}} \frac{q_k}{|\,\mathbf{r}_i - \mathbf{R}_k\,|} \\ &\quad + \sum_\alpha^{\text{QM nuclei}} \sum_k^{\text{point charges}} \frac{Z_\alpha q_k}{|\,\mathbf{R}_\alpha - \mathbf{R}_k\,|}\end{aligned} \quad [4]$$

where $\widehat{H}_0$ designates the gas-phase Hamiltonian of the QM system. The converged SCF energy obtained by the QM program therefore includes *both* the energy of the QM system alone *and* its interaction with the point charges, that is

$$\begin{aligned}E_{SCF} &= \langle \Psi_{QM/MM} | \widehat{H}_0 + \widehat{H}^{el}_{QM-MM} | \Psi_{QM/MM} \rangle \\ &= E_{QM} + E^{el}_{QM-MM}\end{aligned} \quad [5]$$

With the corresponding polarized wave function $\Psi_{QM/MM}$, we can write the energy of the QM system in its environment (but *without* the interaction with the environment) as

$$E_{QM}(\mathbf{R}_{QM};\mathbf{R}_{MM}) = \langle \Psi_{QM/MM} | \widehat{H}_0 | \Psi_{QM/MM} \rangle \quad [6]$$

To evaluate E_{QM} with a standard QM program, we can perform one single SCF cycle using $\widehat{H}_0$ and the polarized wave function.

Free energies from QM/MM molecular-dynamics simulations

Various free-energy methods (such as thermodynamic integration, slow growth, fast growth, free-energy perturbation, or umbrella sampling) have long been developed and used within the framework of classical MD simulations.[14–19] In principle, all these approaches can be applied with a QM/MM, rather than an MM, potential-energy function. An excellent overview of ab initio QM/MM free-energy calculations has recently been given by Hu and Yang.[9] Most commonly, QM/MM free-energy calculations are used to compute the potential of mean force (PMF) along a distinguished coordinate. Usually, the system is driven along this one-dimensional reaction coordinate (RC) from the initial to the final state using constraints (thermodynamic integration, slow growth, fast growth) or restraints (umbrella sampling). The free-energy difference is then obtained by integrating the mean force of constraint or by reconstructing the probability distribution along the reaction coordinate, respectively.

Although feasible in principle, the extended sampling required to produce converged free energies makes these approaches rather expensive when implemented with a QM/MM potential. With a semi-empirical QM method, full QM/MM sampling is well-affordable, as shown here for the chorismate mutase case or in other work from our laboratory.[20–23] However, when a first-principles (ab initio or density functional theory) QM method is required to accurately treat the reactive process, full QM/MM MD simulations become highly computationally demanding, if not prohibitive, even by modern standards (but see, e.g., ref. 24 for an ab initio QM/MM MD study). To alleviate the computational cost of such simulations, methods have been devised that avoid the sampling of the QM region; among them, QM/MM FEP has been widely used. Recent free-energy techniques even allow the computation of the minimum free-energy path and the optimization of stationary points on the free-energy surface.[25–28] QM/MM has also been combined with metadynamics[29–31] as a sampling technique.[32]

QM/MM free-energy perturbation

Here, we first summarize our implementation of the QM/MM FEP method[23] and discuss related schemes thereafter. QM/MM FEP is based on Zwanzig's perturbation formula[33] that expresses the free-energy difference between two states a and b as the exponential average of the potential-energy change between a and b, the (canonical ensemble) average being taken over configurations representative of state a

$$\Delta A^{a\to b} = -k_B T \ln\langle \exp\{-\Delta E^{a\to b}/(k_B T)\}\rangle^a \quad [7]$$

where k_B is Boltzmann's constant and T the temperature. Note that this is not a perturbation treatment in the usual sense; in the limit of complete sampling, eq. [7] is exact. By exchanging the labels of the states, one obtains a "backward" perturbation $\Delta A^{b\to a} = -\Delta A^{a\to b}$.

The basic idea in QM/MM FEP is to use a sequence of QM/MM-optimized structures, typically along a reaction coordinate, as the states and to keep the QM part fixed in each state. The free-energy difference between two states (a, b) along the reaction coordinate is then evaluated by sampling, at the MM level, the energy difference between the QM part being in configuration a to being in configuration b

$$\Delta A^{a\to b}_{env} = -k_B T \ln\langle \exp\{-\Delta E^{a\to b}_{pert}/(k_B T)\}\rangle^a_{MM} \quad [8]$$

with

$$\begin{aligned}\Delta E^{a\to b}_{pert} &= E_{QM-MM}\left(\mathbf{R}^b_{QM}, \mathbf{R}_{MM}(t)\right) \\ &\quad - E_{QM-MM}\left(\mathbf{R}^a_{QM}, \mathbf{R}_{MM}(t)\right)\end{aligned} \quad [9]$$

E_{QM-MM} is the QM–MM interaction energy as defined in eq. [3]; $\mathbf{R}_{MM}(t)$ is the instantaneous configuration of the MM atoms; $\mathbf{R}^a_{QM}$ and $\mathbf{R}^b_{QM}$ are the configurations of the QM atoms fixed in state a and b, respectively.

The subscript "env" in eq. [8] is a reminder that the free-energy difference thus obtained only accounts for the environment and the QM–MM coupling. The free-energy change due to the fixed QM part can simply be approximated in a first step by the corresponding potential-energy difference

$$\Delta A_{QM} \approx \Delta E_{QM} \quad [10]$$

where the QM energy is as defined in eq. [6]. If the two states are stationary points, one can apply the harmonic-oscillator approximation to the QM part to calculate the zero-point vibrational energy, ΔE^{ZPE}_{QM}; the thermal contributions to the internal energy, ΔU^{th}_{QM}; and the entropy contribution, $-T\Delta S_{QM}$:

$$[11] \qquad \Delta A_{QM} = \Delta E_{QM} + \Delta E_{QM}^{ZPE} + \Delta U_{QM}^{th} - T\Delta S_{QM}$$

The final free-energy difference between two stationary points thus becomes

$$[12] \qquad \Delta A = \Delta A_{env} + \Delta A_{QM}$$

Keeping the QM part fixed during the sampling is the first crucial approximation that makes QM/MM FEP computationally viable. The second key approximation is made when calculating the electrostatic QM–MM term. Within the electrostatic embedding scheme, where the QM–MM electrostatic interaction is calculated at the QM level, each change in the configuration of the MM atoms in principle triggers a QM calculation (i.e., a full SCF) to adapt the QM electron density to the new MM positions. We can eliminate the need for these QM calculations in two steps: Firstly, we introduce the approximation that the density will not change when the MM atoms move (frozen-density approximation). The QM–MM electrostatic interaction is then still calculated at the QM level but requires only one SCF cycle. Secondly, we approximate the (frozen) QM density by a point-charge model. Specifically, at each point along the reaction coordinate, we fit a set of point charges, placed on the QM atoms, to reproduce the electrostatic potential of the true density in the vicinity of the QM region (ESP charges). The QM–MM electrostatic interaction, E_{QM-MM}^{el}, during the sampling is thus approximated by the interaction of the ESP charges with the surrounding MM charges, $E_{QM-MM}^{el,ESP}$.

Note that we apply the ESP charge model not only in the calculation of $\Delta E_{pert}^{a\to b}$ but also to evaluate E_{QM} as

$$[13] \qquad E_{QM} = E_{SCF} - E_{QM-MM}^{el,ESP}$$

(cf. eqs. [5] and [6]). This not only eliminates the need for additional SCF cycles altogether but also leads to a consistent description of the QM–MM electrostatic interaction throughout the procedure.

Using the ESP charges, all terms in $\Delta E_{pert}^{a\to b}$ can therefore be evaluated at the classical level, which makes the sampling highly computationally efficient. The computational bottleneck of the procedure is shifted from the sampling to the QM/MM optimizations required to determine the points along the reaction coordinate. In our present implementation, we calculate $\Delta E_{pert}^{a\to b}$ in each MD step (i.e., typically every 1 fs). However, because MD data are correlated, it would actually be sufficient to evaluate $\Delta E_{pert}^{a\to b}$ less frequently as it is done in the related QTCP approach, e.g., every 100 to 1000 MD steps (0.1 to 1 ps).

We validated the different approximations involved in QM/MM FEP using the activation energy of the OH transfer in the enzyme *p*-hydroxybenzoate hydroxylase (PHBH) at the AM1/GROMOS level.[23] As shown in Table 1, the two methods using full QM/MM MD sampling, thermodynamic integration, and umbrella sampling, agree with each other very well and can therefore be considered as a reference. FEP with a full SCF in each sampling step to evaluate E_{QM-MM}^{el} provides a free-energy barrier that is 4–5 kJ mol^{-1} higher. This difference is statistically significant and reflects the approximations introduced by keeping the QM atoms fixed:[1] The dynamics of the environment is decoupled from the dynamics of the QM region.[2] The QM region is not treated dynamically but within the harmonic approximation. The further approximations of the QM/MM FEP approach that make the evaluation of E_{QM-MM}^{el} viable (the frozen-density approximation and the representation of the density by ESP charges) have almost no effect on the result, which differs from FEP with full SCF by 2 kJ mol^{-1}, which is barely statistically significant. We can therefore conclude that the approximations of the QM/MM FEP method lead to deviations in the computed free energies from the fully sampled reference in the order of a few kJ mol^{-1}, which is very acceptable given the massive reduction in computational effort. The major contribution to the deviation stems from keeping the QM region frozen; representing the density by point-charges does not significantly affect the result.

Table 1. Activation free energy of the OH transfer in the enzyme PHBH calculated at the AM1/GROMOS level using different free-energy methods.

Method	$\Delta^{\ddagger}A$ (kJ mol^{-1})[b]	Reference
Thermodynamic integration	101	20
Umbrella sampling	102	23
FEP (full SCF)[a]	106	23
FEP (frozen density)[a]	108	23
FEP (ESP charges)[a]	108	23

[a]The $\Delta^{\ddagger}A$ values from FEP include the QM thermal and entropic terms ($\Delta U_{QM}^{th} - T\Delta S_{QM}$) obtained within the harmonic approximation for the QM part, but are *exclusive* of the zero-point vibrational energy (ΔE_{QM}^{ZPE}) to make them comparable to the thermodynamic-integration and umbrella-sampling results.

[b]The statistical uncertainty of $\Delta^{\ddagger}A$ is 1–2 kJ mol^{-1}.

We also implemented and tested different schemes for handling the link atoms in the FEP approach and found that their effect on the results is small.[23] Moreover, we verified the validity of the ESP fit for densities described by extended basis sets.

Related FEP methods

The QM/MM FEP method described in the previous section is essentially an implementation of the QM/MM-FE method introduced by Zhang et al.,[34] with which it shares all its key features. Rod and Ryde[35,36] presented a closely related approach called QTCP (quantum thermodynamic-cycle perturbation). However, they derived it from slightly different assumptions. They start from a subtractive QM/MM scheme with mechanical embedding, where the QM atoms bear some type of point charges to begin with and the QM–MM electrostatic interaction is described at the classical level. To improve upon this, they introduce two additional FEP calculations (the vertical legs of their thermodynamic cycle) at the initial and final states that recover the free-energy difference due to changing the charge model of the QM region from point charges to a polarizable density. That is, they additionally sample perturbations, in our notation, of the type

$$[14] \qquad \begin{aligned} \Delta E_{pert}^{MM\to QM/MM} &= E_{SCF}\left(\mathbf{R}_{QM}^{a}, \mathbf{R}_{MM}(t)\right) \\ &\quad - E_{QM-MM}^{el,classical}\left(\mathbf{R}_{QM}^{a}, \mathbf{R}_{MM}(t)\right) \end{aligned}$$

(see eq. [16] in ref. 36). $E_{QM-MM}^{el,classical}$ is the QM–MM electro-

static interaction calculated using any kind of point-charge model, not necessarily ESP charges, for the QM region.

This full QTCP scheme is obviously rather costly because of the QM sampling required to evaluate the vertical legs of the cycle. Rod and Ryde therefore introduce approximations akin to the ones used in the QM/MM FEP approach, which eliminate the sampling for the vertical legs: The QM density is kept frozen during the sampling, and the density is represented by ESP charges derived from a QM/MM calculation of the initial state. With these approximations, QTCP essentially becomes equivalent to QM/MM FEP or QM/MM-FE, respectively.

The idea of the QTCP approach to use the FEP formalism to calculate the free-energy due to a change in computational level at the same configuration, rather than between two configurations, was previously exploited by Warshel and co-workers[37] and has been applied also by others.[38–40]

System setups and computational details

Fluorinase

System overview

The enzyme 5′-fluoro-5′-deoxyadenosine synthase (FDAS or fluorinase for short) from the soil bacterium *Streptomyces cattleya* is the only native enzyme known to date to catalyze the formation of C–F bonds.[41] It uses fluoride and *S*-adenosine-L-methionine (SAM) as substrates to produce 5′-fluoro-5′-deoxyadenosine (FDA); see Scheme 1. The enzyme is a hexamer in solution, consisting of a dimer of tightly bound C_3-symmetric homotrimers; each monomer is a chain of 299 amino-acid residues. The trimer unit features three equivalent active sites at the monomer interfaces.

We have previously investigated the C–F bond formation in fluorinase in detail using QM/MM optimization methods at the DFT/CHARMM level.[42] The active site with bound SAM and fluoride is shown in Fig. 2. The position of the fluoride, which is not present in the experimental structure, was determined from classical MD simulations and QM/MM optimizations. In agreement with experimental evidence,[43] we found that the attack of the fluoride at the C5′ position of SAM follows an S_N2-type mechanism. Our study revealed interesting details about the involvement of active-site residues in fluoride binding and produced reaction and activation energies.

Herein, we will report reaction and activation energies for the reaction with the other halides (Cl^-, Br^-, I^-) and free reaction and activation energies for the complete halide series.

System setup

The preparation and setup of fluorinase system was described in detail in ref. 42. In summary, we used a complete trimer unit and hydrated one of the three equivalent active sites with a sphere of water of 25 Å radius centered on the sulfur atom of SAM, which amounts to a total of 17 596 atoms. Inside the hydrated sphere, we defined an active region in which the atoms were free to move while the remaining atoms were kept frozen. To be consistent with previous work,[42] the active region for the optimization of reactant, transition, and product states for the other halides was a shell of 8 Å around SAM and the halide (1394 atoms). For the reaction-profile optimizations and the FEP simulations, the active region was a sphere of 17 Å radius containing 2947 atoms.

Scheme 1. C–F bond formation catalyzed by fluorinase.

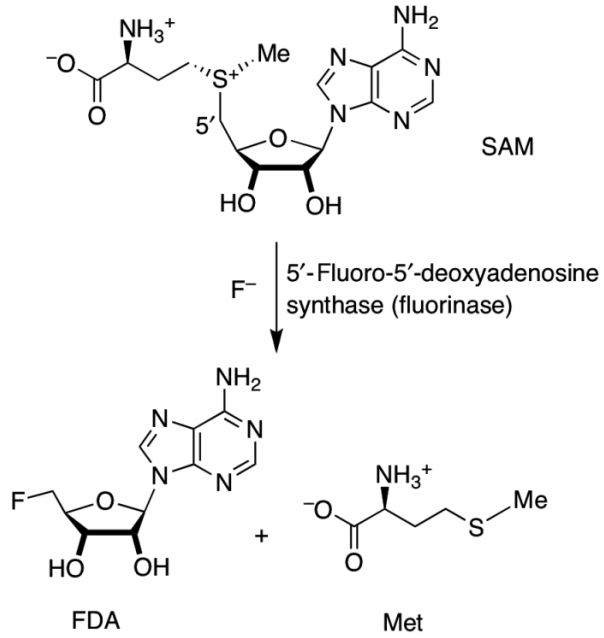
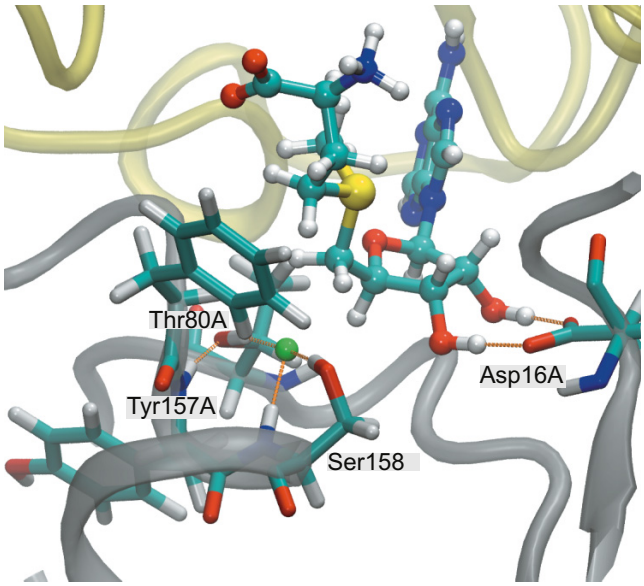


Fig. 2. QM/MM optimized structure of the active site of fluorinase. SAM is shown in ball-and-stick representation; fluoride as a green ball; selected active site residues as sticks; and hydrogen bonds are indicated in orange. The protein chains of the two monomers that form the active site are sketched in grey and yellow, respectively.



The QM/MM calculations were carried out at the DFT(BP86)/CHARMM level with the modular program package ChemShell.[13,44] QM energy and gradient were provided by TURBOMOLE,[45–49] interfaced to ChemShell. The resolution of the identity (RI) approximation[50–53] was employed with the BP86 functional.[54–57] The def2-TZVP[58,59] basis set was used, augmented with one diffuse function on all atoms (TZVP+; see Supplementary data);[41] the core

electrons of iodine were described by an effective core potential (def2-ecp)[60]. MM energy and gradient were evaluated by ChemShell's internal force-field driver using the CHARMM22[61] all-atom topology and parameter data; the MM electrostatic interactions were fully calculated. The QM electron density was electrostatically embedded into the field of the rigid MM point charges by including them into the QM Hamiltonian. The charge-shift scheme was applied at the QM/MM boundary,[13] and no electrostatic QM/MM cutoff was employed. The QM part contained SAM without the 2-ammoniopropanoate moiety, that is, 5′-(*S*,*S*-dimethylsulfonio)-5′-deoxyadenosine, the covalent bond across the QM/MM boundary being saturated with a H link atom (39 QM atoms).

QM/MM geometry optimizations were performed with a linearly scaling algorithm in hybrid delocalized coordinates (HDLCOpt)[62] as implemented in ChemShell. The residues for the optimization algorithm were the standard amino-acid residues. In microiterative transition-state searches, the core region, for which the Hessean is calculated, included the halide, C(5′)H_2, and S; full details can be found in the Supporting Information to ref. 42.

Structure optimizations with other halides

Starting from the reactant, transition, and product states obtained for the native fluoride,[42] we optimized the respective states for chloride, bromide, and iodide by replacing the halide and re-optimizing the active region. This procedure allowed the local environment to adjust to the different sizes and charge distributions of the halides, but it does not account for longer-range effects.

QM/MM reaction profile and harmonic frequencies

The reaction coordinate was defined as the difference between the bond being broken and the bond being formed, $\xi = d(\text{S–C5}) - d(\text{C5′–X})$, where X = F, Cl, Br, I. The reaction profiles were calculated by QM/MM optimizations from $\xi = -1.0$ to 2.2 Å (full profile, for F) or $\xi = -1.4$ to 0.4 Å (profiles from reactant to transition state only, for Cl, Br, I) initially in steps of 0.2 Å. If the FEP forward and backward simulations at a given RC step showed convergence problems, the step size was reduced by inserting additional optimization points. This was only necessary in certain parts of the profile between the transition and the product state. In this region of the RC, the fluoride anion gradually becomes a fluoro substituent bearing a much smaller charge, which leads to a rearrangement of the hydrogen bonding around the fluorine.

To calculate the QM contribution to the free energy at the stationary points from the harmonic approximation, numerical second derivatives were calculated at these points for the QM region (partial Hessean including the QM atoms and the MM boundary atom, consistent with link-atom method 4)[23] using ChemShell. Two displacements of 0.01 au were done per degree of freedom; the SCF convergence was tightened for the Hessean calculations (scfconv 10, grid m4).

Free-energy perturbations

For each QM/MM-optimized structure along the RC, a set of ESP charges for the QM atoms was derived by fitting to the electrostatic potential at the positions of the 200 MM atoms nearest to the QM region. With the QM region (plus the MM boundary atom) held fixed, the active region of the system was then equilibrated by QM/MM MD at 300 K under *NVT* conditions. The QM–MM electrostatic interaction in all simulations was evaluated at the classical level using the ESP charges. The MD time step was 1 fs. The Berendsen thermostat[63] ($\tau_T = 0.1$ ps) was used during the first 10 ps of the equilibration phase; the Nosé–Hoover chain thermostat[64–67] (with a chain length of 4; $\tau_{NHC} = 0.02$ ps, corresponding to $\tilde{\nu}_{NHC} = 375$ cm^{-1}) thereafter. The total equilibration time was 50 ps at each RC point.

The FEP simulations were then carried out; both forward and backward perturbations were done. The sampling time per step was 20 ps, of which the first ps was always discarded. The data of the remaining 19 ps were subjected to coarse-graining and statistical tests as described in ref. 20. Additional data were discarded as necessary to obtain a series with trend-free mean and variance. Note that the sampling length effectively used to calculate the average therefore varies between steps. The maximum deviation between forward and backward perturbation at a given RC step was about 3 kJ mol^{-1}; typical deviations were <1 kJ mol^{-1} (Fig. 3).

The discontinuities in the sampled free-energy increments in Fig. 3 are caused by non-continuous changes between the optimized structures along the reaction path that are not explicitly included in the RC. For example, the dip at $\xi = 0.7$ Å is related to the breaking of a hydrogen bond and the formation of a new one in the vicinity of the fluoride ion in the MM part. This hydrogen-bond rearrangement causes the decrease in A_{env} (plotted in Fig. 3), which is, however, counterbalanced by an increase in E_{QM}. Therefore, the sum $A(\xi) = E_{QM} + A_{env}$ is smooth (see Fig. 7). Such discontinuous changes in the environment free energy that arise from shifts between the QM energy and the environment energy due to structural rearrangements in the MM region do not disappear with longer sampling or smaller steps in ξ; note that the associated error bars are not larger than those for other points along the reaction coordinate.

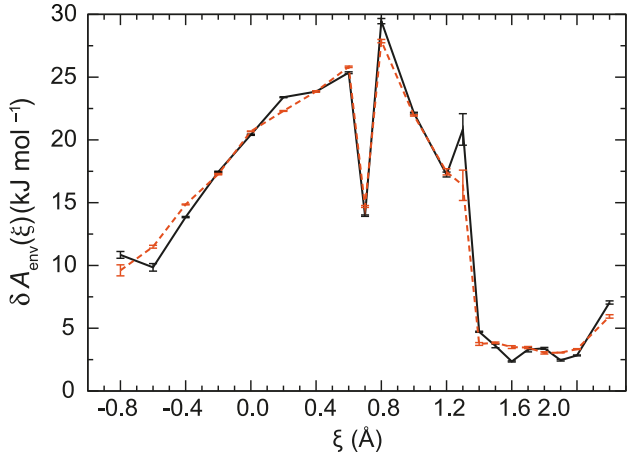
On the other hand, there are features like the one at $\xi = 1.3$ Å (with a larger sampling error and a larger difference between forward and backward perturbation relative to other points) that are related to structural rearrangements in the QM part. Their error bars would be reduced by adding more steps in ξ. In the present case, this was not done at $\xi = 1.3$ Å because the error bar was still deemed acceptable and because the point does not contribute to the barrier, which is of primary interest here.

P450cam

System overview

The cytochrome P450 enzymes constitute a superfamily of haem monooxygenases that perform a variety of essential functions, such as detoxification and biosynthesis, in nearly all living species. They catalyze a variety of oxidation processes, including aliphatic and aromatic hydroxylation, epoxidation, and heteroatom oxidation. The extensive theoretical work on these enzymes has been reviewed.[68] QM/MM studies in our laboratory[69–84] (see ref. 4 for a summary) have focused on P450cam from the bacterium *Pseudomonas putida*, which catalyzes the selective hydroxylation of camphor at

Fig. 3. FEP-sampled environment free-energy increments, $\delta A_{env}(\xi_i \rightarrow \xi_{i+1})$ and $\delta A_{env}(\xi_{i+1} \rightarrow \xi_i)$ for the forward and backward perturbations, respectively, along the reaction coordinate for the fluorination reaction in fluorinase. The increments are plotted at ξ_{i+1} for both sampling directions; hence, there is no value at $\xi_1 = -1.0$ Å. The error bars show the statistical uncertainty (standard deviation of sampled mean). Black line: Forward perturbations; red dashed line: backward perturbations.



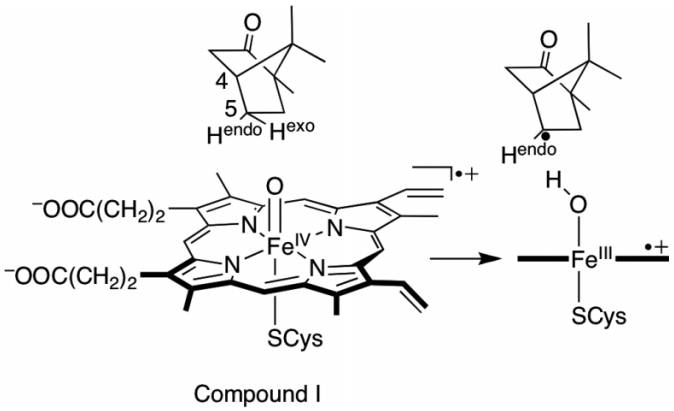
C5 to the exo alcohol (Scheme 2). The crucial reactive intermediate is an iron(IV)-oxo species (Compound I) with three unpaired electrons that give rise to almost degenerate doublet and quartet states. Compound I is able to abstract an aliphatic hydrogen from the substrate. Subsequently, the substrate radical thus generated recombines with the iron(III) hydroxo ligand to form the hydroxylated product.

We have previously studied the hydrogen-abstraction step in P450cam at the DFT(B3LYP)/CHARMM level by QM/MM optimizations.[71,77,78,80] Herein, we report on FEP simulations of this step.

System setup

We started from an optimized hydrogen-abstraction transition state obtained previously. The details of preparing and setting up the system have been published previously.[71,77,78,80] The procedures used were conceptually and technically very similar to the ones described above for fluorinase. We only highlight system-specific differences from here on. The enzyme was fully hydrated by enclosing it into a water shell of 15 Å, the outer 10 Å of which were kept fixed; the total system size is about 24 000 atoms. The active region contained 444 atoms in optimizations and 7439 atoms for the FEP simulations. The QM region included 51 atoms (region R1):[71,77,78,80] the oxy-haem without the side chains; the thiolate S of the distal cysteine; and the $C(4)HC(5)H_2C(6)H_2$ fragment of camphor. Twelve hydrogen link atoms were used to saturate the bonds across the QM–MM boundary; the QM region therefore was [FeO(porphine)(SH)] + propane. The electronic structure was described by the B3LYP functional[54,55,85–88] as implemented in TURBOMOLE. The basis set of double-ζ quality denoted B1[71,77,78,80] was used. All calculations were done in the quartet state ($S = 3/2$). The MM region was described by the CHARMM22[61] force field. All QM/MM calculations were done with ChemShell.

Scheme 2. Hydrogen abstraction from camphor by Compound I of P450cam.



QM/MM reaction profile and FEP simulations

The reaction coordinate was defined as the difference between the bond being broken and the bond being formed, $\xi = d(C5–H^{exo}) - d(H^{exo}–O)$. Starting from the optimized transition state, the reaction profile was produced initially from $\xi = -1.2$ to 1.4 Å in steps of 0.2 Å and subsequently refined to a step size of 0.1 Å. Frequency calculations at the stationary points were not performed for this case.

At the transition state, the system was equilibrated for 30 ps at 300 K (time step of 1 fs) with the QM part and MM boundary atoms being frozen in addition to the outer water layer (7388 active atoms). For the other points along the RC, the QM region of the equilibrated TS was replaced by the QM/MM-optimized QM region of the respective point and then re-equilibrated for 20 ps. A cutoff of 15.9 Å was used for the non-bonded MM–MM interactions with a pair-list updated every 25 steps. No QM/MM cut-off was applied.

For the FEP simulations, both forward and backward perturbations were performed. The sampling time at each RC point was 20 ps. Only the trend-free[20] part (no less than 8 ps) of the data was used to calculate the average. The P450cam system is particular in that it has rather many link atoms.[12] The impact of different link-atom treatments is therefore more pronounced. With steps of 0.2 Å and link-atom method 4,[23] where both QM and MM boundary atoms and the link atoms are perturbed, the sampled free-energy differences deviated between the forward and backward perturbation by 19 kJ mol^{-1} on average, which was considered unacceptable. Applying link-atom method 2,[23] where the MM boundary atoms remain unperturbed, the average deviation between the forward and backward perturbation was still 15 kJ mol^{-1}. Only by reducing the step size to 0.1 Å and using link-atom method 4, we obtained an acceptable agreement (maximum deviation of 7 kJ mol^{-1}) between forward and backward perturbations. Only these results will be discussed below.

Chorismate mutase

System overview

The enzyme chorismate mutase catalyzes the Claisen rearrangement of chorismate to prephenate, without covalent-

bond formation between the substrate and the enzyme (Scheme 3). The reaction, a rare example of a pericyclic reaction in biochemistry, is a key step of the shikimate pathway for the synthesis of aromatic amino acids in plants, fungi, and bacteria. It has been the subject of numerous theoretical studies devoted to the structure of the enzyme–substrate complex, reaction pathways, and the role of specific residues in transition-state stabilization. It has also served as an important case for testing and developing theories on the origin of enzymatic catalysis. A summary of recent theoretical work is provided in ref. 4. Here, we have investigated the Claisen rearrangement in the *Bacillus subtilis* chorismate mutase (BsCM) by means of umbrella-sampling free-energy simulations at the semi-empirical QM/MM level.

Preparation and setup

Full details about system preparation and setup, methods, and programs can be found in the Supplementary data. All simulations were performed with the program CHARMM[89,90] using the self-consistent-charge density-functional tight-binding (SCC-DFTB) method[91,92] for the QM region (substrate, 24 atoms) and the CHARMM22[61] all-atom force field for the MM region (protein and solvent atoms, 13 397 atoms). No electrostatic QM/MM cut-off was employed; MM electrostatic interactions were treated with the extended electrostatics method.[90,93]

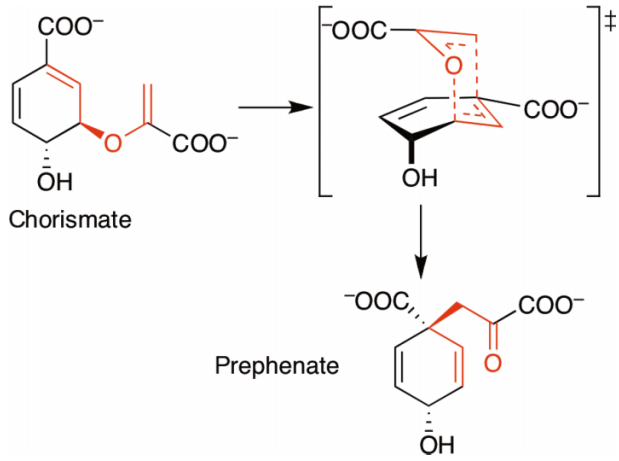
The initial coordinates of BsCM bound to a transition-state analogue (TSA) were obtained from the Protein Data Bank (PDB code 2CHT). The crystal structure[94] consists of four trimers in the asymmetric unit; only the first one was retained. The TSA in the active site at the interface between chains A and B was converted to chorismate; the other two TSA molecules were removed. The BsCM–chorismate complex was hydrated with a 30 Å sphere of water. Protein and substrate atoms within 17 Å from the origin of the sphere and all (crystallographic and added) water molecules within 30 Å were free to move (active region). The system was then subjected to QM/MM MD simulations under *NVT* conditions (time step 1 fs). After 200 ps of equilibration, 10 snapshots were taken every 10 ps as initial configurations for the subsequent QM/MM simulations. Depending on the snapshot, the active region included about 9850 atoms.

Umbrella-sampling simulations

The free energy of activation and the free energy of reaction of chorismate to prephenate in the active site of BsCM were obtained from umbrella-sampling[95] simulations along a reaction coordinate defined as the difference between the lengths of the breaking C–O and the forming C–C bond. The biased MD simulations consisted of two parts: First, the snapshots were equilibrated in the reactant state (ξ = –2.5 Å) at 300 K for 20 ps. Then, the RC was sequentially changed from –2.5 Å to 2.5 Å in steps of 0.1 Å (51 windows). In each window, the system was sampled for 10 ps, and RC data were collected in every time step. Only the data of the last 9 ps were used for the analysis. The sampling was then repeated in the backward direction starting from ξ = 2.5 Å.

We used umbrella integration (UI)[21,22] as well as the weighted histogram analysis method (WHAM)[96] to combine the windows of the US simulations. The number of bins was always chosen to be 500. Using 1000 bins left the free-energy results virtually unchanged for either UI or WHAM. UI and WHAM yielded essentially identical results, differing by no more than 0.8 kJ mol^{-1}. The root-mean-square deviation (RMSD) between UI and WHAM results was 0.4 kJ mol^{-1} for the activation free energies and 0.2 kJ mol^{-1} for the reaction free energies. The values reported below were obtained using UI.

Scheme 3. Claisen (allyl vinyl ether to 4,5-unsaturated ketone) rearrangement of chorismate to prephenate in chorismate mutase.



Potential-energy profiles

Complementary to the free-energy reaction profiles obtained from US, we also calculated potential-energy profiles for all the snapshots. The structures were energy-minimized and then subjected to reaction-path calculations using the same RC definition as for the US simulations. The reaction pathways were calculated in both the forward and backward direction. To precisely locate minima and maxima in the reaction profiles, the energy curves were fitted by cubic splines.

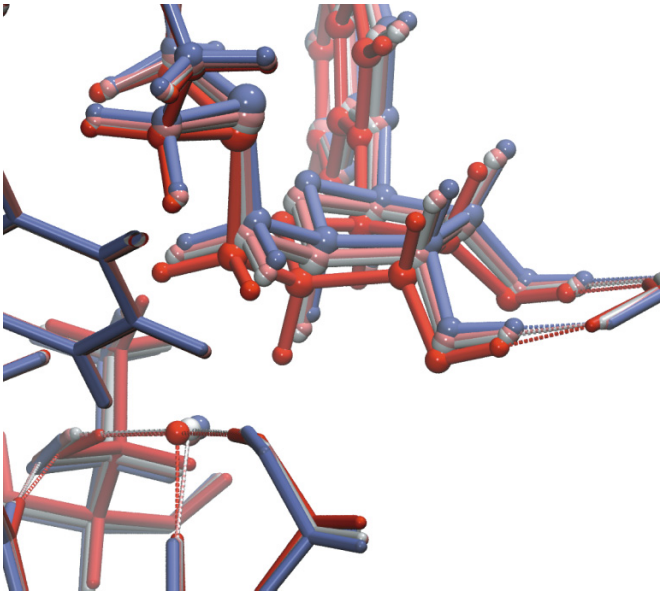
Results

Fluorinase

Optimizations with halides other than fluoride

The QM/MM-optimized reactant states with bound SAM and halide for all four halides are shown in Fig. 4. It can be seen that the structures are largely superimposable and only minor adjustments have taken place (here and below, structure were superimposed by aligning all fixed atoms). The structures with Cl^- and Br^- in particular are nearly identical. While the chosen optimization procedure does not allow for larger rearrangements, the stable binding position of SAM is explained by the highly specific recognition in the binding site through multiple hydrogen bonds with the adenine part and salt bridges and hydrogen bonds to the zwitterionic amino acid of the methionyl part. Additionally, the ribose part is held in position by the double hydrogen bond to Asp 16A. The halide binding site is in all cases formed by hydrogen bonds from the side-chain OH and the backbone NH of Ser 158A and the side-chain OH of Thr 80A. Selected structural parameters of the reactant states are collected in Table 2. The trends seen along the halide series follow the expected pattern. With increasing size of the halide, and

Fig. 4. Superimposed optimized reactant states for the complete halide series in the fluorinase: F (red), Cl (white), Br (salmon), and I (blue); for clarity, the halides are all represented by a sphere of the same size.



thus decreasing charge density, the hydrogen-bonding distances to the halide become monotonically longer. The biggest changes are seen from F to Cl and then from Br to I, while the Cl and Br systems are very similar. The distance between the halide and the attacked C5′ also increases along the series. However, only the change from F to Cl is larger than the other two. The slightly lengthened C5′–S bond for F compared with the other halides suggests an increased degree of activation towards nucleophilic attack.

The optimized transition states are presented in Table 3 and Fig. 5. The pertinent changes between reactant and transition state have already been discussed for fluoride[42] and apply also to the other halides, i.e., the C5′–S bond is essentially broken, the X–C5′ bond is partially formed, and the coordination around C5′ is trigonal-planar; in brief, it is a nearly ideal S_N2 transition state. As the reaction progresses, the attacking halide's negative charge is increasingly delocalized over the substrate as the interaction between halide and substrate increases. This is reflected in weakened hydrogen bonds to the halide, especially the Thr 80A OH···X bond is starting to break at the transition state. The hydrogen bonds from Ser 158A, however, are only slightly lengthened at this stage, somewhat more for F than for the other halides. In the product states (Table 4 and Fig. 6), the X–C5′ bond is fully formed and the methionine leaving group is well-separated from the halogenated deoxynucleoside. Only the hydrogen bond from the Ser 158 amide NH to the halogen is retained. Remarkably, the halogen position does not differ very much along the series, despite the differences in C5′–X bond lengths (the C–I bond is 0.7 Å longer than the C–F bond). The RMSD from F is between 0.1 Å (Br) and 0.4 Å (I); the ribose unit as a whole is displaced relative to the F system by 0.3 Å (Cl, Br) to 0.5 Å (I). To accommodate the longer C5′–X bonds for the heavier halogens, while keeping the halogen itself almost at the same position, the exocyclic torsion about C4′–C5′ needs to adjust. There are

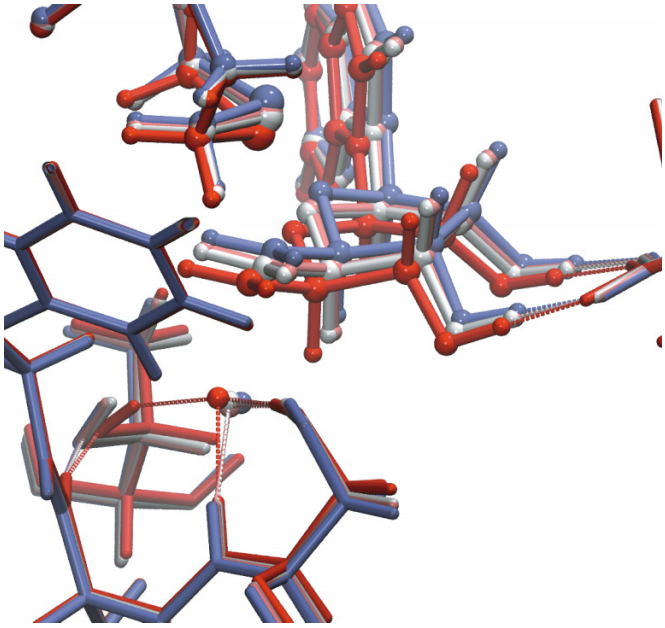
Table 2. Selected bond lengths, hydrogen-bonding distances, and RC values (in Å) in the optimized reactant states (RS) in the fluorinase for all the halides.

	F	Cl	Br	I
d(C5′, S)	1.85	1.84	1.83	1.83
d(X, C5′)	2.76	2.99	3.12	3.24
ξ_{RS} (opt)	–0.91	–1.15	–1.29	–1.41
d(Ser158A OH···X)	1.79	2.26	2.32	2.58
d(Ser158A NH···X)	1.82	2.14	2.16	2.40
d(Thr80 OH···X)	1.82	2.25	2.29	2.54

Table 3. Selected bond lengths, hydrogen-bonding distances, and RC values (in Å) in the optimized transition states (TS) in the fluorinase for all the halides.

	F	Cl	Br	I
d(C5′, S)	2.47	2.53	2.57	2.51
d(X, C5′)	1.93	2.23	2.34	2.50
ξ_{TS} (opt)	0.54	0.30	0.23	0.01
d(Ser158A OH···X)	1.85	2.30	2.34	2.59
d(Ser158A NH···X)	1.92	2.17	2.19	2.42
d(Thr80 OH···X)	2.06	2.45	2.54	2.72

Fig. 5. Superimposed optimized transition states for the complete halide series in fluorinase: F (red), Cl (white), Br (salmon), and I (blue); for clarity, the halides are all represented by a sphere of the same size.



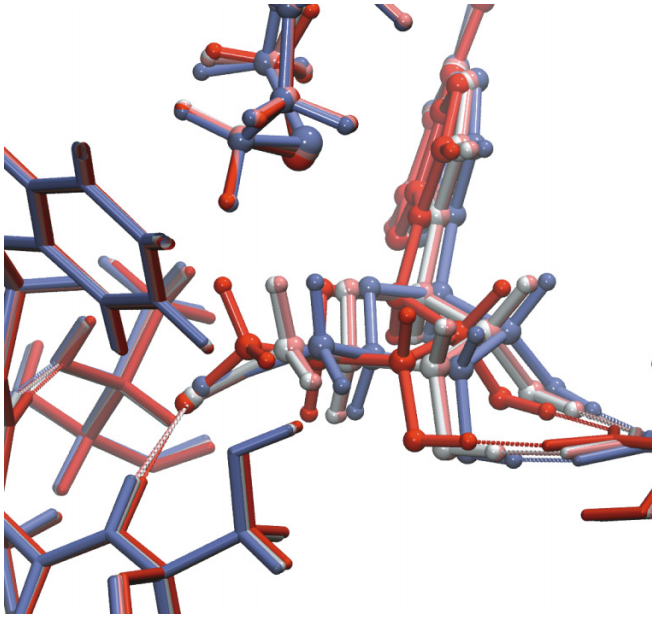
experimental structures available of the fluoro-[41] and the chloro-product.[97] Essentially, they agree perfectly with the QM/MM-optimized structures, which validates our approach.

The energetic data from the optimizations of the stationary points are summarized in Table 5. They reveal three main results: (*i*) The reaction is exothermic only with fluoride, while it is endothermic with the other halides; (*ii*) the forward barrier is lowest for fluoride; (*iii*) therefore, the backward barriers for the other halides are significantly

Table 4. Selected bond lengths, hydrogen-bonding distances, RC values (in Å), and torsion angles η about C4′–C5′ in the optimized product states (PS) in fluorinase for all the halides.

	F	Cl	Br	I
d(C5′, S)	3.44	3.45	3.44	3.47
d(X, C5′)	1.43	1.80	1.95	2.13
ξ_{PS} (opt)	2.01	1.65	1.49	1.34
d(Ser158A NH···X)	2.00	2.27	2.28	2.50
η(O4′, C4′, C5′, X)	100.2°	93.7°	93.5°	87.6°

Fig. 6. Superimposed optimized product states for the complete halide series in fluorinase: F (red), Cl (white), Br (salmon), and I (blue); for clarity, the halogens are all represented by a sphere of the same size.



lower than the backward barrier for fluoride. Fluorinase thus has a clear preference in the C–X bond-forming step for fluorination over other halogenations. This is also reflected in the extent to which the C–X bond has formed in the transition state (Table 5), which, in agreement with the Hammond postulate, shows an earlier transition state for fluoride than for the other halides.

The calculated preference for fluoride over chloride is in agreement with the experimental data:[97] Fluorinase was found to act also as a chlorinase, however, with the equilibrium of the reaction being shifted to the reactant side and with much lower activity, as expected from the calculated reaction and activation energies. The measured ratio of effective rate constants of the forward reactions with fluoride and chloride corresponds to a difference in activation energy ($\Delta\Delta^{\ddagger}G$ at 298 K) of 12 kJ mol^{-1}. The excellent agreement with the calculated difference of 10 kJ mol^{-1} is certainly fortuitous; however, it again provides support for the validity of our approach in the present case.

Experimentally, no activity has been observed with bromide or iodide. Our results would not suggest such a discrimination between, e.g., chloride and bromide, although the reaction with bromide is more endothermic and has a somewhat higher barrier. However, it should be kept in mind that we have investigated only the C–X bond-forming step, while the halide selectivity may well be determined earlier in the mechanism. For instance, the halide-binding step, which will involve the (partial) replacement of the halide's hydration water by interactions with the protein, may well be decisive for the halide selectivity.[98,99] It is interesting to note that the recently reported SAM chlorinase SalL,[100] which is structurally, and thus presumably also mechanistically, similar to fluorinase, accepts chloride, bromide, and iodide, but not fluoride, as halide substrate.

Table 5. Reaction and activation energies (in kJ mol^{-1}) as well as relative lengths of the forming and breaking bonds in the TS for C–X bond formation in fluorinase. Data from QM/MM optimizations of stationary points (at BP86/def2-TZVP+/CHARMM level).

	F	Cl	Br	I
ΔE	–22	16	35	22
$\Delta^{\ddagger}E$(fw)	59	69	74	65
$\Delta^{\ddagger}E$(bw)	81	53	39	43
d(X–C5′)$_{TS}$/d(X–C5′)$_{PS}$	1.35	1.24	1.20	1.17
d(C5′–S)$_{TS}$/d(C5′–S)$_{RS}$	1.34	1.38	1.40	1.37

Free energies

The forward activation free energies obtained from the FEP simulations for C–X bond formation in fluorinase with X = F, Cl, Br, and I are collected in Table 6. We also provide the contributions of the individual components that sum up to the total free-energy difference according to eqs. [11] and [12]. With fluoride only, we determined the complete free-energy profile from reactants to products (Fig. 7), which allowed us to calculate also the reaction free energy: $\Delta_r A$ = –19.5 ± 0.6 kJ mol^{-1} (composed of $\Delta_r A_{env}$ = 261.9 ± 0.6 kJ mol^{-1}, $\Delta_r E_{QM}$ = –275.1 kJ mol^{-1}, $\Delta_r E_{QM}^{ZPE}$ = 0.0 kJ mol^{-1}, and $\Delta_r U_{QM}^{th} - T\Delta_r S_{QM}$ = –6.3 kJ mol^{-1}).

Comparing the free energies (excluding ΔE_{QM}^{ZPE}) with the corresponding potential energies from Table 5, the main conclusion is that the effect of finite temperature on the reaction and activation energies investigated is small (<3 kJ mol^{-1}) in this case. Without the ZPE and finite-temperature contributions to the free energy from to the QM part, the potential-energy and free-energy profiles are essentially identical from the reactant up to the transition state (shown for fluoride in Fig. 7). Also the values of the reaction coordinate at the stationary points of the free-energy curve differ only insignificantly from their potential-energy counterparts.

An explanation for this small difference between potential and free energies in the reactive step can be found in the high degree of pre-organization and pre-orientation provided by the active site. The reactants are ideally set up and held in place for the S_N2 process by their various interactions with the protein throughout the reaction. Moreover, only minimal adjustments to the active-site structure are necessary to accommodate either the reactants or the products. There is therefore not much room for thermal and entropic contributions to the free energy.

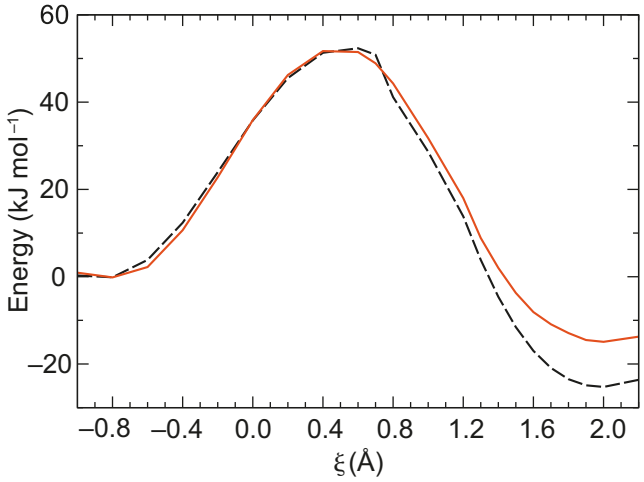
Cytochrome P450cam

The results for the hydrogen abstraction from camphor by Compound I of cytochrome P450cam are summarized in Table 7. Like for fluorinase, the differences between poten-

Table 6. Activation free energy (for the forward reaction, $\Delta^{\ddagger}A$(fw), in kJ mol^{-1}) and its components according to eqs. [11] and [12] from QM/MM FEP for C–X bond formation in fluorinase. Also given are the values of the reaction coordinate (in Å) at the stationary points of the free-energy profile.

	F	Cl	Br	I
$\Delta^{\ddagger}A$(fw)	53.5±0.6	64.5±1.7	71.2±0.3	61.5±1.0
$\Delta^{\ddagger}A_{env}$	123.2±0.6	110.4±1.7	118.2±0.3	103.4±1.0
$\Delta^{\ddagger}E_{QM}$	–64.5	–40.1	–40.5	–36.1
$\Delta^{\ddagger}E_{QM}^{ZPE}$	–4.6	–4.8	–5.3	–5.1
$\Delta^{\ddagger}U_{QM}^{th} - T\Delta^{\ddagger}S_{QM}$	–0.6	–1.0	–1.2	–0.7
ξ_{RS} (FEP)	–0.84	–1.15	–1.25	–1.41
ξ_{TS} (FEP)	0.51	0.28	0.23	0.02

Fig. 7. Reaction profile of the C–F bond formation in fluorinase. Dashed line: Potential-energy profile; solid line: free-energy profile. Note that the free-energy curve only includes $\Delta A_{env} + \Delta E_{QM}$ because the harmonic approximation for the calculation of the QM ZPE and finite-temperature terms is valid only at stationary points.



tial and free energies are small (see Fig. 8), although we note that the free-energy values here were calculated as $\Delta A_{env} + \Delta E_{QM}$ only (i.e., not including the ZPE and finite-temperature contributions from the QM part). Previous gas-phase model calculations at the B3LYP level[66] gave ZPE and enthalpic finite-temperature corrections to the QM barrier of –18 and +2 kJ mol^{-1}, respectively. Thus, the overall finite-temperature contribution remains small (nominally 4 kJ mol^{-1}, ignoring $\Delta^{\ddagger}S_{QM}$), while the ZPE correction to the barrier is significant for hydrogen transfer and should be applied to both ΔE and ΔA. The values of the reaction coordinate at the stationary points differ only slightly between the potential-energy and free-energy profiles, respectively: ξ_{RS} = –0.99/–1.03 Å (opt/FEP); ξ_{TS} = 0.17/0.18 Å; ξ_{PS} = 1.10/1.07 Å. Again, the small effect of finite temperature on structure and energetics of the hydrogen-abstraction reaction can be attributed to the pre-orientation of the bound reactants. Figure 9 shows the optimized TS; camphor is seen to be hydrogen-bonded to an active-site Tyr residue.

Chorismate mutase

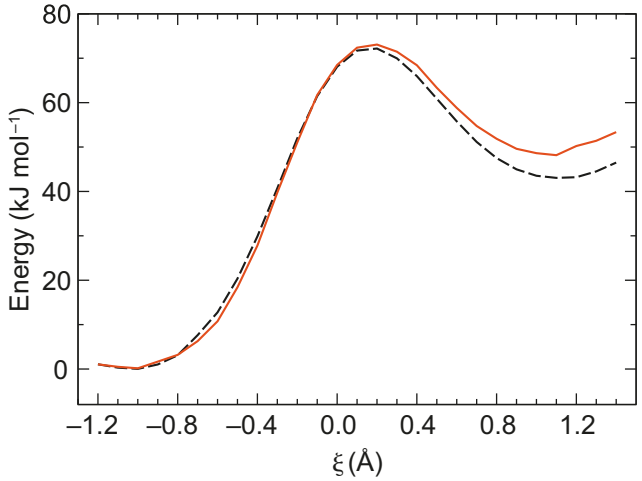
Table 8 reports the activation and reaction energies obtained by optimizations for the conversion of chorismate to

Table 7. Activation and reaction energies (in kJ mol^{-1}) for the hydrogen abstraction in P450cam. Both potential energies and free energies are given.

	Forward barrier	Reaction energy
ΔE(opt)	71	42
ΔA(FEP)	73±2	48±2
ΔA_{env}	–15±2	–19±2
ΔE_{QM}	88	67

Note: Free energies do not include the QM ZPE and finite-temperature terms.

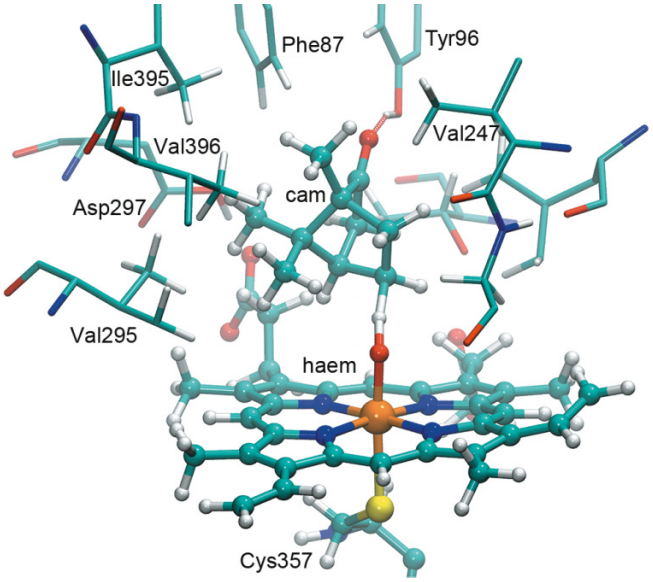
Fig. 8. Reaction profile of the H-abstraction in P450cam. Dashed line: Potential-energy profile; solid line: free-energy profile. Note that the free-energy curve only includes $\Delta A_{env} + \Delta E_{QM}$.



prephenate in *B. subtilis* chorismate mutase. The values for the barrier range from 21.8 to 38.7 kJ mol^{-1} across the forward and backward scans of the 10 snapshots, with a mean of $\Delta^{\ddagger}E$ = 27.5 ± 1.2 kJ mol^{-1} (the error bar indicated is the single standard deviation of the mean over the 20 values, corresponding to a confidence level of 68%). The reaction energies cover the interval from –68.5 to –98.4 kJ mol^{-1}; the mean is $\Delta_r E$ = –83.5 ± 1.9 kJ mol^{-1}. The standard deviations of the individual energy values are 5.3 kJ mol^{-1} and 8.3 kJ mol^{-1}, respectively. The difference between forward and backward scan for a given snapshot is at most 1.0 and 0.5 kJ mol^{-1} for $\Delta^{\ddagger}E$ and $\Delta_r E$, respectively. The associated RMSD (forward vs. backward) is 0.3 kJ mol^{-1} for the barriers and 0.2 kJ mol^{-1} for the reaction energies.

Multiple potential-energy profiles for the reaction of interest were previously calculated with hybrid DFT QM/MM methods (B3LYP/6–31G(d)/CHARMM27).[101] From 16 different reaction pathways, an average reaction barrier of 50.2 kJ mol^{-1} was obtained, with a standard deviation of 7.1 kJ mol^{-1}. Like in the present work, a large spread of the calculated barriers from 37.7 to 62.8 kJ mol^{-1} was reported. The differences in barrier height between the 16 QM/MM reaction profiles were explained by differences in the microstructure of the active site. Another research group[102] performed reaction-path calculations for the rearrangement of chorismate to prephenate in BsCM at various QM/MM levels of theory. At the B3LYP/6–31+G(d)/CHARMM22 level, they calculated $\Delta^{\ddagger}E$ = 37.4 kJ mol^{-1} and $\Delta_r E$ = –81.6 kJ

Fig. 9. Transition-state structure for the hydrogen abstraction from camphor in cytochrome P450cam. Camphor and haem are represented as all-atom ball-and-stick models. The residues of the binding pocket are shown as sticks; only hydrogen atoms involved in hydrophobic interactions or H-bonds (Tyr) with camphor are shown. Colour code: C cyan, H white, N blue, O red, S yellow, Fe orange.



mol^{-1}, while their best estimates (34.3 and –96.7 kJ mol^{-1}, respectively) were obtained using MP2 instead of B3LYP. We also refer to ref. 102 for an overview of previous theoretical results. The experimental value[103] for the activation enthalpy is $\Delta^{\ddagger}H$ = 53.1 ± 1.7 kJ mol^{-1} (at 298 K). Comparing the calculated value for $\Delta^{\ddagger}E$ to the experimental result for $\Delta^{\ddagger}H$, as is usually done in the literature,[101] the experimental barrier is clearly underestimated by the present value of $\Delta^{\ddagger}E$ = 27.5 ± 1.2 kJ mol^{-1}; the relative error is –48%.

The Helmholtz free energies of activation and reaction for the Claisen rearrangement of chorismate to prephenate in BsCM calculated with US for the 10 snapshots are shown in Table 9. It should be noted that free energies from US (and other techniques where the full system is sampled by classical mechanics) do not include any ZPE contributions. The $\Delta^{\ddagger}A$ values from forward and backward runs range from 30.8 to 46.1 kJ mol^{-1}, with a mean of 36.5 ± 0.8 kJ mol^{-1}, which is thus our best estimate for the activation free energy. The $\Delta_r A$ values lie within a range of –67.6 to –88.6 kJ mol^{-1}; the mean is –75.2 ± 1.3 kJ mol^{-1}. The standard deviation of the individual values amounts to 3.5 and 5.7 kJ mol^{-1} for the $\Delta^{\ddagger}A$ and $\Delta_r A$ data, respectively. We note that propagating the statistical sampling errors reported in Table 9 would produce smaller error bars for the mean (0.3 and 0.5 kJ mol^{-1}, respectively). However, these would not properly represent the distribution of the individual values.

The free-energy barriers obtained from the forward and backward runs on a given snapshot agree within 5 kJ mol^{-1}; the RMSD is 4.4 kJ mol^{-1}. The 30 ps and 40 ps snapshots are exceptions where the deviation is larger (6.7 and 9.4 kJ mol^{-1} in absolute value, respectively). For the reaction free energies, four snapshots (10 ps, 40 ps, 80 ps, and 90 ps) have a forward-versus-backward difference of >5 kJ mol^{-1} in absolute value (9.5, 7.7, 13.5, and 11.6 kJ mol^{-1}, respectively). The RMSD is 7.0 kJ mol^{-1}. The variability of values between sampling directions for the same snapshot as well as between different snapshots may be explained as a consequence of yet incomplete sampling. The distribution could be narrowed by equilibrating more extensively in each window and (or) additional sampling.[20]

Table 8. Potential energies of activation and of reaction (in kJ mol^{-1}) for the conversion of chorismate to prephenate in chorismate mutase.

	Forward scan		Backward scan	
Snapshot (ps)	$\Delta^{\ddagger}E$	$\Delta_r E$	$\Delta^{\ddagger}E$	$\Delta_r E$
10	26.0	–80.4	26.0	–80.5
20	22.6	–98.4	22.6	–98.3
30	23.4	–92.0	22.4	–92.5
40	38.7	–76.3	38.5	–76.6
50	30.7	–85.7	30.7	–85.7
60	21.8	–87.3	21.8	–87.3
70	31.7	–76.0	31.8	–75.9
80	22.6	–83.4	22.5	–83.6
90	28.4	–68.5	28.4	–68.6
100	29.8	–85.9	29.8	–86.1

Table 9. Free energies of activation and of reaction (in kJ mol^{-1}) obtained with US for the conversion of chorismate to prephenate in chorismate mutase.

	Forward sampling		Backward sampling	
Snapshot (ps)	$\Delta^{\ddagger}A$	$\Delta_r A$	$\Delta^{\ddagger}A$	$\Delta_r A$
10	36.8±1.2	–75.1±2.3	33.9±1.3	–84.6±2.2
20	38.3±1.2	–73.3±2.1	34.6±1.2	–75.7±2.1
30	40.8±1.3	–69.1±2.1	34.1±1.1	–73.2±2.0
40	36.7±1.4	–75.3±2.3	46.1±1.3	–67.6±2.2
50	32.8±1.2	–73.1±2.2	34.6±1.1	–74.5±2.1
60	38.5±1.3	–74.3±2.3	40.5±1.3	–74.8±2.3
70	35.2±1.3	–73.0±2.4	34.2±1.4	–74.6±2.3
80	34.5±1.3	–74.0±2.2	30.8±1.3	–87.5±2.2
90	36.8±1.2	–77.0±2.3	33.3±1.3	–88.6±2.3
100	37.6±1.4	–68.4±2.3	39.0±1.3	–69.3±2.2

Note: US free energies do not include ZPE contributions.

The present result for the free energy of activation of 36.5 ± 0.8 kJ mol^{-1} agrees well with the value of 33.5 kJ mol^{-1} obtained from multiple-steering MD simulations employing a DFT QM/MM Hamiltonian.[104] In the same study, the reaction free energy was calculated to be about –88 kJ mol^{-1}, which lies at the lower end of the present range of $\Delta_r A$ values. Recently,[105] the free-energy profile was calculated by ab initio QM/MM combined with constrained MD-FEP simulations, yielding 41.4 kJ mol^{-1} for the free-energy barrier and about –84 kJ mol^{-1} for the reaction free energy. The experimental value[103] for the Gibbs free energy of activation is 64.4 kJ mol^{-1} (298 K). All the calculations above thus clearly underestimate the barrier. The relative error for the present free-energy barrier (–43%) is similar to the one of the activation enthalpy (–48%).

The overall finite-temperature effect on the barrier, calculated from our best estimates for $\Delta^{\ddagger}A$ and $\Delta^{\ddagger}E$ (see above), is $T\Delta^{\ddagger}S - \Delta^{\ddagger}U_{th}$ = –9.0 ± 1.4 kJ mol^{-1} at T = 300 K. It is dominated by the entropic contribution, since the change of the thermal contribution to the internal energy, $\Delta^{\ddagger}U_{th}$, is

generally very small. Alternatively, we can calculate individual values for the finite-temperature effect on the barrier for all the snapshots (forward and backward direction) reported in Tables 8 and 9. The corresponding 20 values range from –18.7 to +2.0 kJ mol^{-1}, with a mean of –8.9 ± 1.2 kJ mol^{-1} and a standard deviation of 5.6 kJ mol^{-1}. Obviously, both estimates for the finite-temperature effect on the barrier are close to each other (–9.0 ± 1.4 vs. –8.9 ± 1.2 kJ mol^{-1}) and in good agreement with the experimental[103] value of $T\Delta^{\ddagger}S$ = –11.4 ± 1.5 kJ mol^{-1} at T = 298 K. The difference between free and potential energy is thus larger in this case than for the two systems discussed above.

The finite-temperature effect on the reaction energy can be computed analogously either from our best estimates for $\Delta_r A$ and $\Delta_r E$ (see above) or from the corresponding individual snapshot values, which yields –8.3 ± 2.3 and –8.3 ± 2.6 kJ mol^{-1}, respectively. The 20 snapshot values range from –25.1 to +20.0 kJ mol^{-1}, with a standard deviation of 11.6 kJ mol^{-1} (indicating a larger scatter than in the case of the barriers).

Discussion

In this paper, we have presented potential-energy and free-energy data from QM/MM optimizations and MD-based simulations for the carbon–halide bond formation in fluorinase, the hydrogen abstraction from camphor by Compound I in cytochrome P450cam, and the chorismate-to-prephenate rearrangement in *B. subtilis* chorismate mutase. For the purpose of discussion, we complement these results with similar data obtained for the OH transfer in *para*-hydroxybenzoate hydroxylase (PHBH). The free energy of activation in PHBH at the AM1/GROMOS level is 101 kJ mol^{-1} (average from thermodynamic-integration runs on 10 snapshots; individual values range from 95 to 106 kJ mol^{-1}).[20] The potential-energy barriers obtained from geometry optimizations of the corresponding snapshots at the same level vary between 90 and 109 kJ mol^{-1} (avg. 100 kJ mol^{-1}). The finite-temperature contribution to the barrier is thus small for PHBH; nominally (judging from the average values) of the order of 1 kJ mol^{-1}, which is well within the statistical uncertainty of the thermodynamic-integration simulations and the RMS deviation of the results for the different snapshots.

For chorismate mutase, the availability of data from ten different snapshots (with a forward and a backward value each; Tables 8 and 9) enables us to assess the uncertainty of "single-snapshot" results. Statistically, a measurement falls within an interval of ±2σ around the mean with 95% probability, where σ is the sample standard deviation. Treating the 20 data points as statistically independent, one obtains 2σ error bars of 10.6 and 7.0 kJ mol^{-1} for $\Delta^{\ddagger}E$ and $\Delta^{\ddagger}A$, respectively, and of 16.6 and 11.4 kJ mol^{-1} for $\Delta_r E$ and $\Delta_r A$, respectively. Alternatively, one may consider the maximum deviations from the mean, which are 11.2 ($\Delta^{\ddagger}E$), 9.6 ($\Delta^{\ddagger}A$), 15.0 ($\Delta_r E$), and 13.4 ($\Delta_r A$) kJ mol^{-1}. Combining these values with the PHBH snapshot data,[20] we can assign a rather approximate, but practically useful, error bar of about 10 kJ mol^{-1} for (free) energy differences obtained on a single snapshot of a biomolecular system. The errors tend to be smaller for free energies than for potential energies because the former involve sampling. Also, they appear to be somewhat smaller for barriers than for reaction energies.

If one wants to improve on the accuracy of a "single-snapshot" value, be it a free-energy difference obtained from some kind of sampling technique or, even more so, a potential-energy difference obtained by optimizations, it is necessary to perform calculations for several independent snapshots extracted from extended MD simulations. Such a protocol is arguably the most practical way to include effects due to the variability of the wider environment into QM/MM results.

Leaving reaction energies aside, we now focus on activation barriers, in particular on the entropic contributions to barriers. Strictly speaking, the results obtained from umbrella sampling or thermodynamic integration represent $T\Delta S - \Delta U_{th}$, but since ΔU_{th} is generally small (of the order of 1 kJ mol^{-1}), we shall use the term "entropic contribution" for these quantities (as well as for $T\Delta^{\ddagger}S$ itself). We find the calculated entropic contribution (at 300 K unless noted otherwise) to be small for fluorinase, P450cam, and PHBH (between 1 and 4 kJ mol^{-1} in absolute value); only for BsCM, it is somewhat higher (9 kJ mol^{-1}). To put these values into perspective, we have compared to gas-phase data. The Claisen rearrangement in BsCM lends itself particularly well to such a comparison because the reaction is intramolecular, which makes the choice of gas-phase model obvious, and is experimentally known to have a sizeable entropic contribution to the barrier also in the enzyme.

We have calculated the activation free-energy for the rearrangement of chorismate to prephenate in the gas phase at SCC-DFTB level and obtained (at 300 K) $T\Delta^{\ddagger}S$ = –13.6 kJ mol^{-1} ($\Delta^{\ddagger}G$ = 94.0 kJ mol^{-1}; the chorismate reactant was taken to be in the reactive conformation). Compared with our value in the enzyme of –9.0 kJ mol^{-1}, the activation entropy is therefore 50% higher in the gas phase. This substantial reduction of the entropic contribution in the enzyme is attributed to the pre-organization of the reactant in the active site that is found to be significant even for an intramolecular process like the Claisen rearrangement. The reduced, but still fairly large, negative activation entropy in the enzyme is rationalized by considering the tight, rather rigid pericyclic TS. In the chorismate reactant, even when enzyme-bound, the interaction between the two reacting carbon atoms is only weakly preformed and leaves sufficient motional freedom to the fragments, which are only connected by the C–O bond. In the TS, however, both the forming C–C and the breaking C–O interactions are partial covalent bonds, with correspondingly low flexibility with respect to distance and orientation. So, in a case like this, where the reaction has a high (negative) activation entropy related to the intrinsic structure of the TS, the pre-organization of the reactant by the enzyme can only partially compensate for it.

The situation is rather different for the other enzymatic reactions studied herein. They are bimolecular processes and as such inherently entropically disfavoured in the gas phase because of the loss of translational degrees of freedom. For the H-abstraction from camphor by Compound I, $T\Delta^{\ddagger}S$ = –43 kJ mol^{-1} in the gas phase at the B3LYP level.[68] This high entropy contribution is essentially annihilated in the chemical step in the enzyme (to –2 kJ mol^{-1}) owing to the binding and pre-organization of the reactants in the ac-

tive site. The dominating influence of the pre-orientation of the reactants in intermolecular reactions is also illustrated in PHBH. Gas-phase calculations of the barrier with respect to a pre-reactant complex, where the translational degrees of freedom have already been lost, yield $T\Delta^{\ddagger}S$ = –5 kJ mol^{-1} at the AM1 level. In the enzyme, the entropic contribution is further reduced to –1 kJ mol^{-1}, which can again be explained by motionally restricting interactions of the reactants with the protein; in this case, the hydrogen bond of the transferring OH to a proline residue in the TS plays a special role.[20]

For fluorinase, finally, we do not have suitable gas-phase reference data. However, for an intermolecular S_N2 substitution, $T\Delta^{\ddagger}S$ in the gas phase is expected to be strongly negative relative to separated reactants. In the enzyme, most of the entropic contribution should be quenched, which we find indeed to be the case (–0.9 to 2.5 kJ mol^{-1}).

Conclusions

In this contribution, we have reported potential-energy and free-energy data for three enzymatic reactions: carbon–halogen bond formation in fluorinase, hydrogen abstraction from camphor in cytochrome P450cam, and chorismate-to-prephenate Claisen rearrangement in chorismate mutase. The results were obtained by semi-empirical or DFT QM/MM optimizations and two types of QM/MM free-energy simulations (free-energy perturbation and umbrella sampling). On the basis of these results and our previously published free-energy data for PHBH, we have discussed the importance of finite-temperature effects in the chemical step of enzyme reactions. Consistent with the notion that enzymes bind and pre-organize the reactants in the active site, we find that the entropic contribution to the activation barrier tends to be small, particularly when compared with an intermolecular process in the gas phase. In the latter case, the enzyme will "quench" the part of the entropic contribution that is related to the association of the reactants. If a reaction has a transition state that is intrinsically tight (compared with the reactant) due to the specific type of bond-making and -breaking, there may still be a sizeable entropic contribution to the barrier of the reaction in the enzyme, as seen in the chorismate mutase case. Furthermore, it should be emphasized that there will undoubtedly be considerable finite-temperature effects associated with the binding of the reactants preceding the chemical step that are well worth investigating by free-energy techniques. On the other hand, when focusing on localized chemical events in the catalytic cycle of an enzyme, the present evidence suggests that QM/MM optimization studies may often provide barriers that are close to the corresponding free-energy barriers. Our results also highlight the importance of performing biomolecular QM/MM optimizations or free-energy simulations on several snapshots taken from MD simulations if one wants to improve the accuracy of energy differences beyond about 10 kJ mol^{-1}.

Supplementary data

Supplementary data for this article are available on the journal Web site (canjchem.nrc.ca) or may be purchased from the Depository of Unpublished Data, Document Delivery, CISTI, National Research Council Canada, Ottawa, ON K1A 0R6, Canada. DUD 5257. For more information on obtaining material, refer to cisti-icist.nrc-cnrc.gc.ca/cms/unpub_e.shtml.

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